

Source Analysis Report

Introduction

The objective of this project is to develop an Automated Data Collection System capable of extracting recruitment-related information from a publicly accessible online source. The collected information is then stored in a structured database and automated for periodic execution.

For this project, the YC Jobs platform was selected as the primary data source. YC Jobs is a job board maintained by Y Combinator that contains startup job opportunities across various domains such as software engineering, product management, marketing, design, and operations.

The platform provides valuable recruitment information including company names, job titles, locations, job descriptions, and application links, making it suitable for automated data collection and analysis.

Source Selection

Selected Source

YC Jobs

Website Description

YC Jobs is an online recruitment platform where startups associated with Y Combinator publish job openings. The platform serves as a centralized location for job seekers interested in startup careers.

The website is publicly accessible and regularly updated with new job opportunities.

Reasons for Selection

The following factors influenced the selection of YC Jobs:

- Publicly accessible information
 - No account required for browsing job listings
 - Large volume of recruitment-related data
 - Frequently updated content
 - Well-structured presentation of job information
 - Suitable for demonstrating web automation techniques
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Source Investigation

Before developing the scraper, the website was analyzed using browser developer tools.

Investigation Activities

- Examined page structure
- Identified HTML elements containing job information
- Observed dynamic content loading behavior
- Tested accessibility of job listings

- Verified availability of required data fields

Findings

The website uses modern web technologies and dynamically renders content. Traditional static HTML scraping methods may not reliably retrieve all job information.

Therefore, browser automation was determined to be the most appropriate approach.

Data Elements Collected

The following information was identified for collection:

Field	Description
Company Name	Name of hiring company
Job Title	Position offered
Job URL	Direct link to job posting
Location	Job location
Collection Timestamp	Date and time of collection

These fields provide sufficient information for recruitment analysis and future processing.

Technical Challenges

Several challenges were identified during source analysis:

Dynamic Content

The website loads content dynamically, requiring browser automation instead of simple HTTP requests.

Duplicate Records

Some job listings may reappear during multiple collection cycles. Deduplication mechanisms were therefore required.

Structural Changes

Future modifications to website structure could affect scraping logic. Therefore, selectors were chosen carefully to improve maintainability.

Ethical and Legal Considerations

The data collection process follows responsible web scraping practices.

Measures taken include:

- Collecting only publicly available information

- Avoiding excessive request rates
- Respecting website functionality
- Using collected data only for educational and demonstration purposes

No personal or sensitive information is collected.

Conclusion

The source analysis confirmed that YC Jobs is an appropriate platform for demonstrating automated data collection techniques. The website provides relevant recruitment data, public accessibility, and sufficient complexity to showcase browser automation tools.

The analysis phase successfully identified the required data elements, collection challenges, and technical approach required for implementation.